

Unit Conversion

Key Concept in Unit Conversion:
Write the conversion factor as a ratio of 1:
e.g., 1 km = 1000 m is written as $1 = \frac{1000\text{ m}}{1\text{ km}}$ or $1 = \frac{1\text{ km}}{1000\text{ m}}$
Choose the ratio so the units cancel appropriately.

Example 1: Force Conversion
The thrust of a rocket engine is 686 N. Convert this to pounds-force (lbf).
(1 lbf = 4.448 N)

Solution:
 $686\cancel{\text{N}} \times \frac{1\cancel{\text{lbf}}}{4.448\cancel{\text{N}}} = 154\text{ lbf}$

Example 2: Area Conversion
Express an area of 2.40 m² in square centimeters (cm²). (1 m = 100 cm)

Solution:
 $2.40\text{ m}^2 \times \left(\frac{100\cancel{\text{cm}}}{1\cancel{\text{m}}}\right)^2 = 2.40\cancel{\text{m}^2} \times \frac{10,000\cancel{\text{cm}^2}}{1\cancel{\text{m}^2}} = 2.40 \times 10^4\text{ cm}^2$

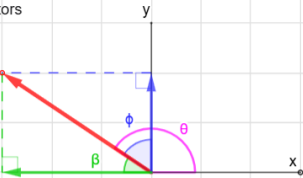
Step

Unit Vectors

Show the labels of vector A and its component vectors

Show θ

y-axis to the left of the x-axis



$|\vec{A}| = A = 3.6056$

$\theta = 146.31^\circ$

$A_x = A \cos \theta = -3.0000$

$A_y = A \sin \theta = 2.0000$

Show the possible angles in a right triangle: $\beta = 33.690^\circ$ $\phi = 56.310^\circ$

$A_x = +A \cos \beta = +A \sin \phi$ for $\vec{A}$ in 1st and 4th quadrants

$A_x = -A \cos \beta = -A \sin \phi$ for $\vec{A}$ in 2nd and 3rd quadrants

$A_y = +A \cos \phi = +A \sin \beta$ for $\vec{A}$ in 1st and 2nd quadrants

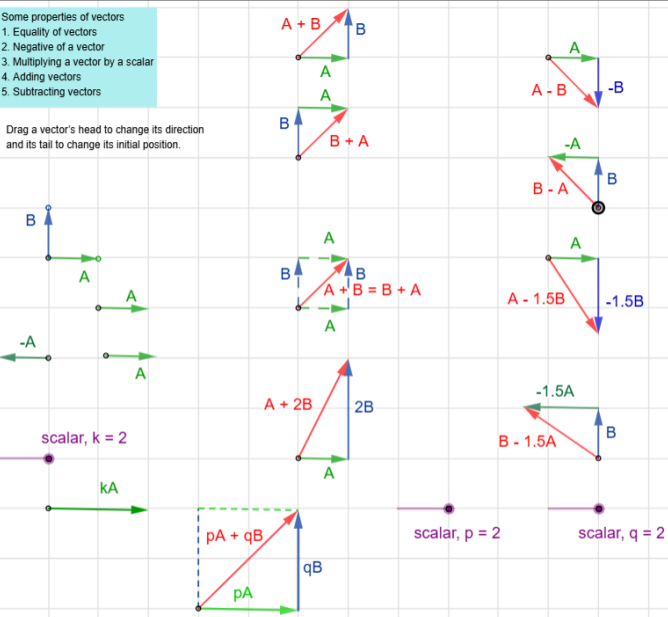
$A_y = -A \cos \phi = -A \sin \beta$ for $\vec{A}$ in 3rd and 4th quadrants

Drag the tip of vector A to change its direction and length.
Drag the x-axis tip to change its orientation.

Some Properties of Vectors

- 1. Equality of vectors
- 2. Negative of a vector
- 3. Multiplying a vector by a scalar
- 4. Adding vectors
- 5. Subtracting vectors

Drag a vector's head to change its direction and its tail to change its initial position.



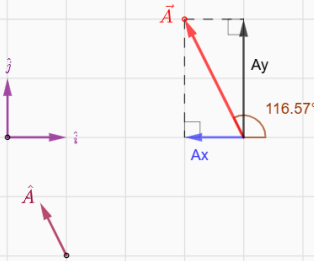
Vector Components

Show component vectors

Show the angle of vector A

$|\vec{A}| = 2.2361$

$\theta = 116.57^\circ$



$$\begin{aligned}\vec{A} &= \vec{A}_x + \vec{A}_y = A_x \hat{i} + A_y \hat{j} \\ &= (-1.0000) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + (2.0000) \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -1.0000 \\ 2.0000 \end{pmatrix} \\ &= -1.0000 \hat{i} + 2.0000 \hat{j}\end{aligned}$$

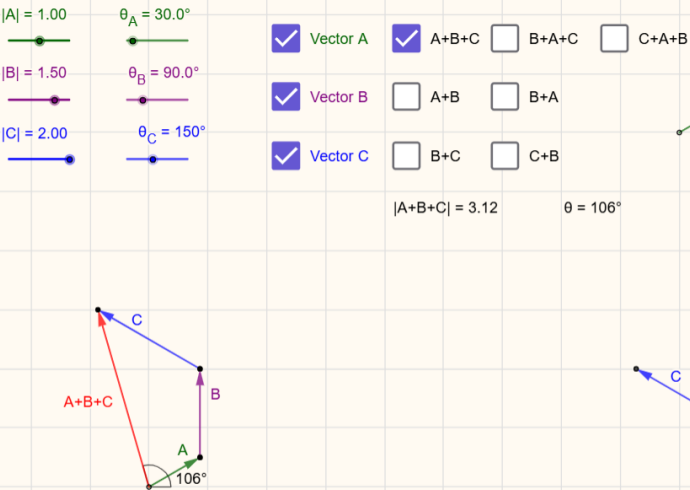
Unit vector, $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$

$\vec{A}_x$ and $\vec{A}_y$ are component vectors of $\vec{A}$

A_x and A_y are x - and y -components of $\vec{A}$

Drag the tip of vector A to change its direction and length.
Changing the length of vector A does not change the length of its unit vector, which is always 1.

Adding 3 Vectors Graphically



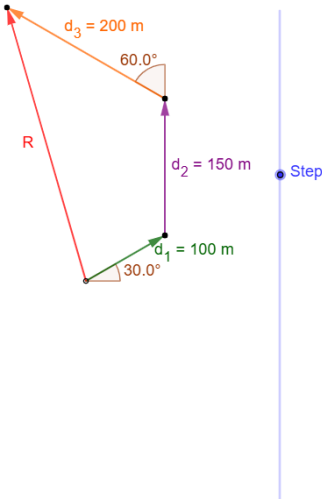
Vector Addition: Worked Example

Example: Displacement of a Moving Object on a 2D Plane
A delivery robot moves on a flat warehouse floor starting from point A. Its path is:
100 m at 30° north of east, then 150 m due north, and finally 200 m at 60° west of north.
Find the resultant displacement of the robot from point A.

Solution:
Step 1: Sketch the displacement vectors
Let east be +x and north be +y.

Step 2: Resolve each displacement into components

$$\vec{d}_1 = d_{1x} \hat{i} + d_{1y} \hat{j} = \begin{pmatrix} 100 \text{ m} \cos 30^\circ \\ 100 \text{ m} \sin 30^\circ \end{pmatrix} = \begin{pmatrix} 86.6 \\ 50 \end{pmatrix} \text{ m}$$



Dot Product (Scalar Product)

Geometric definition: $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$ Algebraic definition: $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$

$$\begin{aligned}\vec{F} \cdot \vec{d} &= |\vec{F}| |\vec{d}| \cos \phi \\ &= (1.4142 \text{ N})(2.0000 \text{ m}) \cos(45.000^\circ) \\ &= 2.0000 \text{ N}\cdot\text{m}\end{aligned}$$

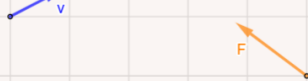
$$\begin{aligned}\vec{F} \cdot \vec{v} &= |\vec{F}| |\vec{v}| \cos \theta \\ &= (1.5000 \text{ N})(1.0000 \text{ m/s}) \cos(115.69^\circ) \\ &= -0.65026 \text{ N}\cdot\text{m/s}\end{aligned}$$

Drag the tip of a vector to change both its magnitude and direction or only its direction.

The scalar product of two vectors is no longer a vector, but a scalar.

The scalar product of two vectors is negative when the angle between them is greater than 90° .

Task: Move the tails of vectors $\vec{F}$ and $\vec{v}$ to the same point to determine the angle between them.



Finding Displacement from x-t and v-t Graphs

$$x(t) = \begin{cases} \text{If } (t \geq 0, t^2 - 4t) \end{cases}$$

$$v(t) = dx/dt = 2t - 4, \quad (t \geq 0)$$

☒ Show position x(t) vs. time t graph

Displacement, $\Delta x = x_f - x_i$
from time a to b = 8 m

Alternatively, from the v-t graph,

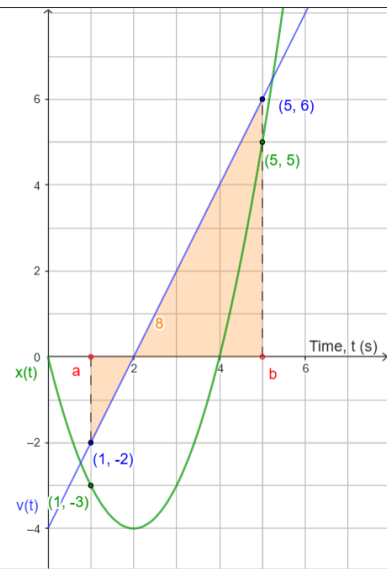
$$\text{displacement, } \Delta x = \int_a^b v(t) dt$$

☒ Show velocity v(t) vs. time t graph

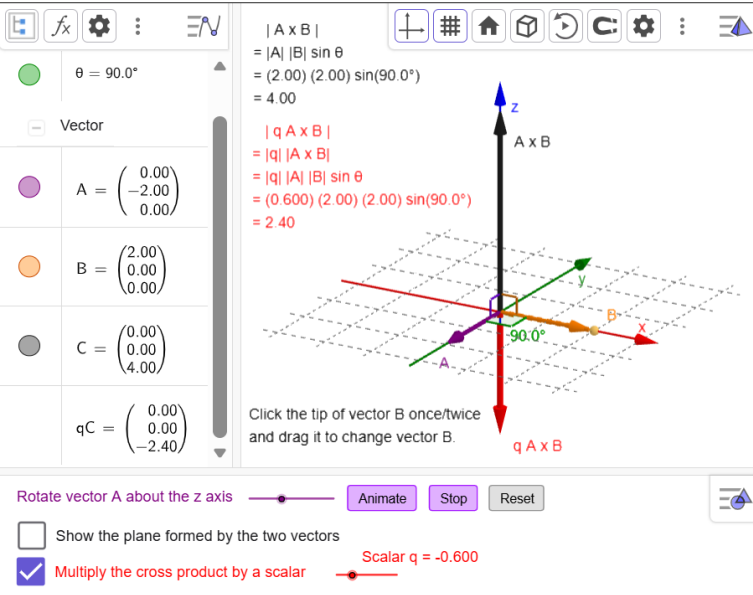
☒ Show area under the v(t) curve

☐ Speed |v(t)| vs. time t graph

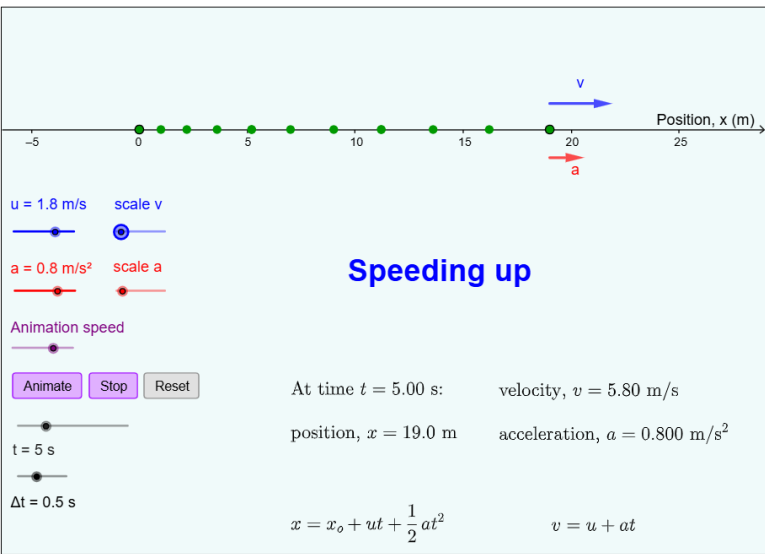
Distance travelled from time a to b = 10 m



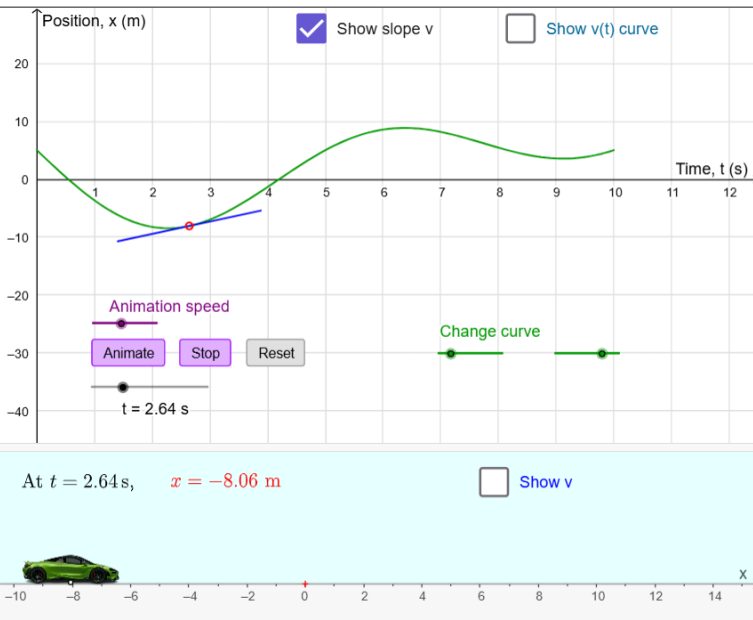
Cross Product (Vector Product)



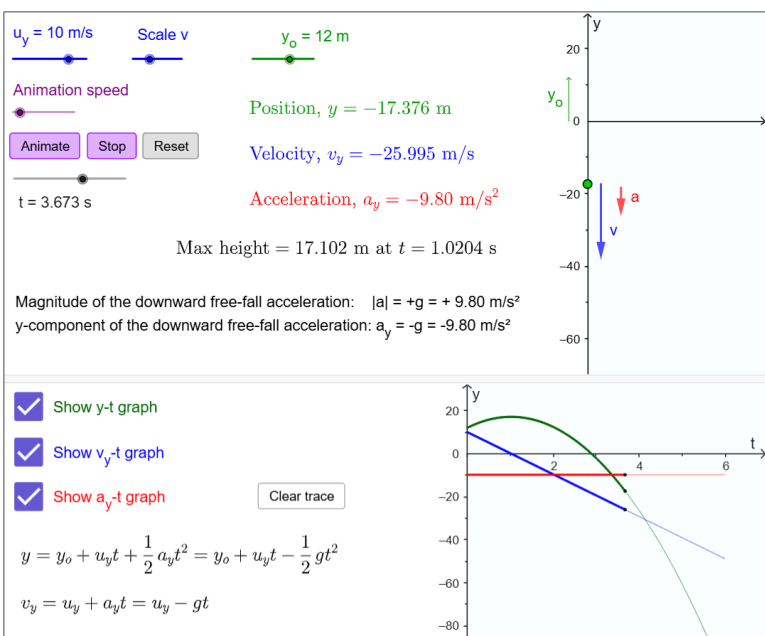
1D Motion with Constant Acceleration (Motion Diagram)



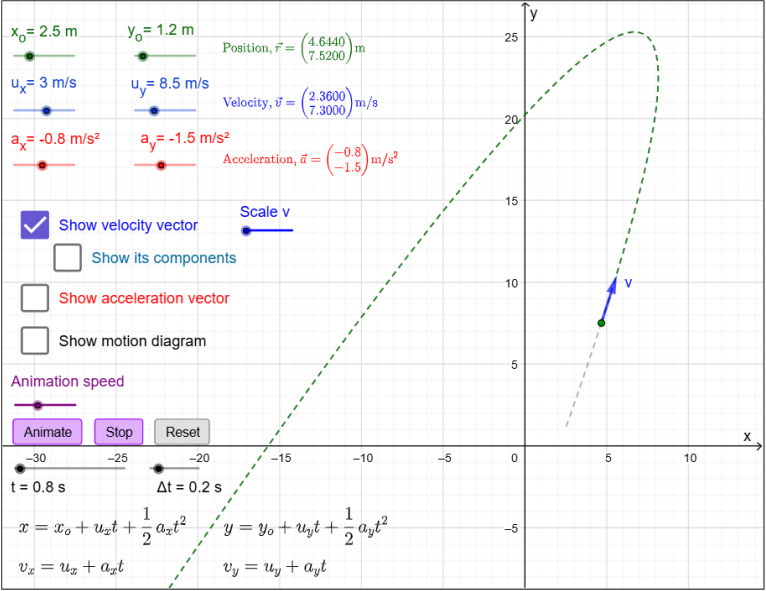
Position and Velocity in 1D Motion



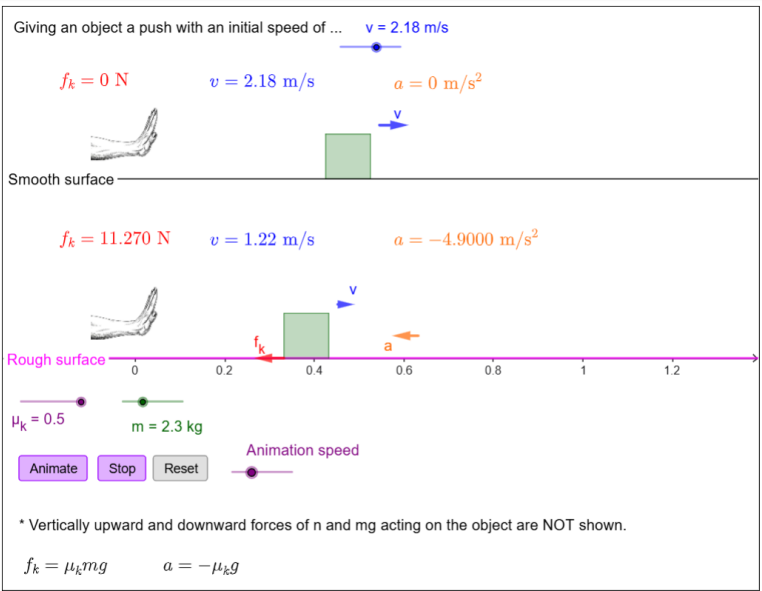
1D Free Falling



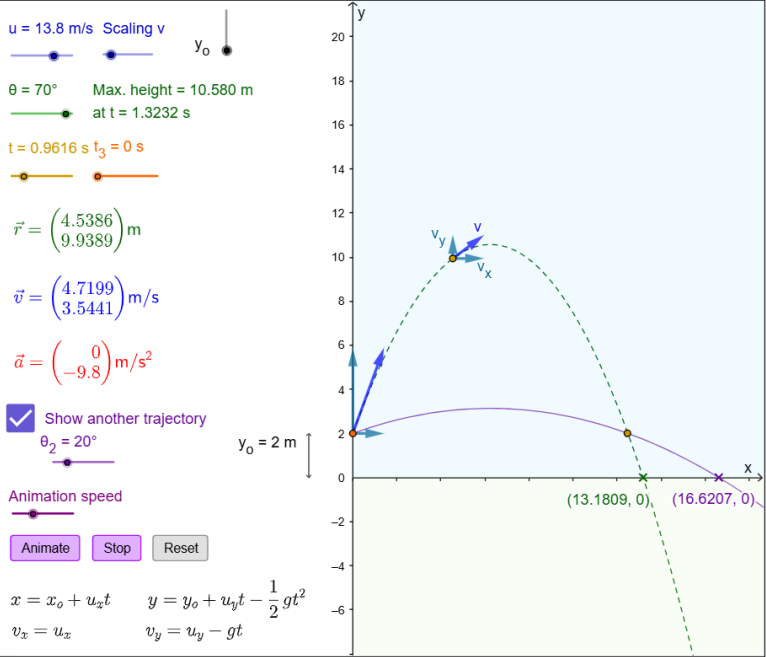
2D Motion with Constant Acceleration



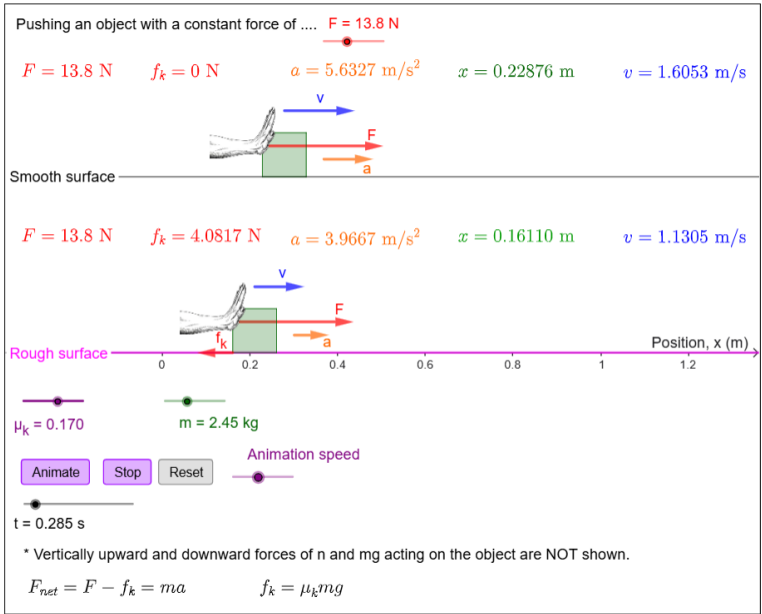
Giving an Object an Initial Push



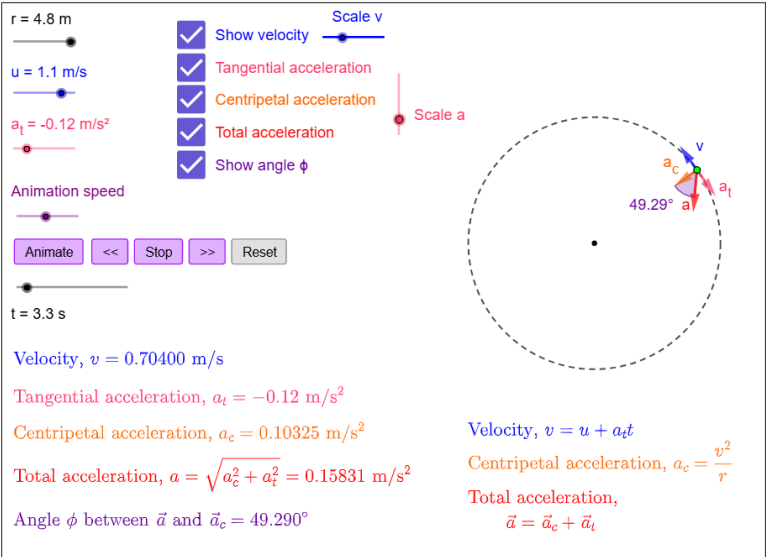
Projectile Motion



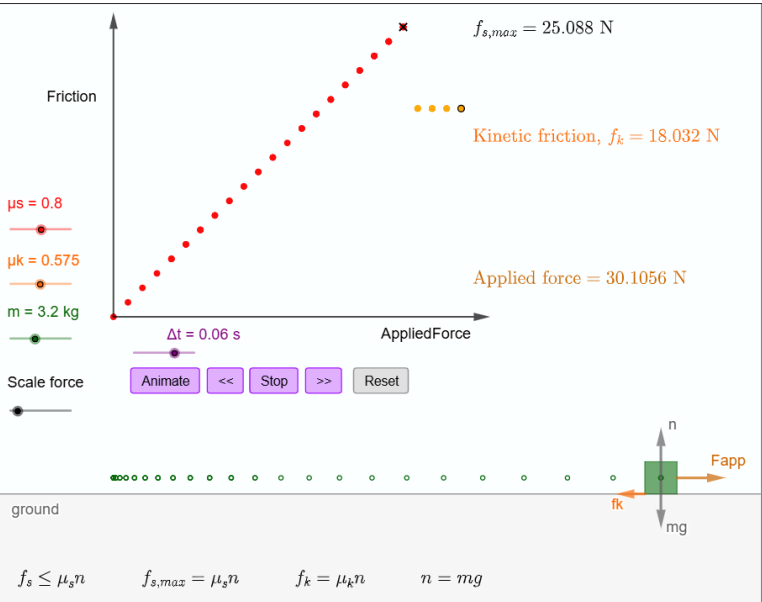
Pushing an Object



Circular Motion



Static Friction and Kinetic Friction



Forces and Accelerations

$m = 2 \text{ kg}$
 $h_0 = 4.45 \text{ m}$
 Animation speed
 Animate Stop Reset
 Show forces ☒ Show accelerations ☐
 $n = 16.00492 \text{ N}$
 $mg = 19.6 \text{ N}$
 $f_k = 0 \text{ N}$
 $v = 4.87244 \text{ m/s}$
 $a_c = 5.33497 \text{ m/s}^2$
 $a_t = 9.42998 \text{ m/s}^2$
 $a = 10.8345 \text{ m/s}^2$
 $\mu = 1.08$
 Rough surface of 4 m
 3.2387430126 m
 Angle of centripetal acceleration from the vertical, $\theta = 74.20522^\circ$
 On the curved surface: $a_c = v^2/r$ $a_t = g \sin \theta$ $n = m(v^2/r + g \cos \theta)$
 On the horizontal surface: $a = \mu_k g$ $f_k = \mu_k n$ $n = mg$

Work done by Friction and Mechanical Energy on an Incline

$m = 4.2 \text{ kg}$
 $u = 8 \text{ m/s}$
 scale v
☒ Show friction
 $\theta = 30^\circ$
 $\mu_k = 0.448$
 $x_0 = 1.02 \text{ m}$
 Box size
 $d = 2.7979 \text{ m}$
 $v = -1.3895 \text{ m/s}$
 Max. incline displacement = 3.6772 m
 Total distance travelled, $d_{\text{total}} = 4.5566 \text{ m}$
 ΔK
 ΔU
 ΔE
 W_f
 $K_i = 134.40 \text{ J}$
 $U_i = 20.992 \text{ J}$
 $E_i = 155.39 \text{ J}$
 $W_f = -72.765 \text{ J}$
 $K = 4.0544 \text{ J}$
 $U = 78.572 \text{ J}$
 $E = 82.627 \text{ J}$
 $\Delta K = -130.35 \text{ J}$
 $\Delta U = 57.581 \text{ J}$
 $\Delta E = -72.765 \text{ J}$
 Animation speed
 Animate Stop Reset
 $t = 2.185 \text{ s}$
☒ Energy bars
 Kinetic energy, $K = \frac{1}{2}mv^2$
 Gravitational potential energy, $U = mgy$
 Total mechanical energy, $E = K + U$
 Work done friction, $W_f = -f_k d_{\text{total}}$

$$= -(\mu_k mg \cos \theta) d_{\text{total}}$$

Energy Transformation on a Curved Ramp and Friction Track

$h_0 = 4.8 \text{ m}$ $m = 2.22 \text{ kg}$
 Animation speed
 [Animate] [Stop] [Reset]

$K = 79.914 \text{ J}$ $v = 8.4850 \text{ m/s}$
 $U = 24.515 \text{ J}$ $y = 1.1268 \text{ m}$
 $E = 104.43 \text{ J}$
 [Work done by friction, $W_f = 0.0000 \text{ J}$]

$\mu = 0.674$ Rough surface of 4.35 m
 [Show friction]

Kinetic energy, $K = \frac{1}{2}mv^2$
 Gravitational potential energy, $U = mgy$
 Total mechanical energy, $E = K + U$
 Work done friction, $W_f = -f_k d = -\mu_k m g d$

Mass-Spring System

$k = 35 \text{ N/m}$
☒ Energy bars
 $A = 0.5 \text{ m}$
 $m = 1 \text{ kg}$

$K = 4.1044 \text{ J}$
 $U = 0.27058 \text{ J}$
 $E = 4.3750 \text{ J}$

$v = 2.865107629 \text{ m/s}$
 $x = -0.1243449436 \text{ m}$

Animation speed

$F_s = -kx$
 $F_s = -(35 \text{ N/m})(-0.12434 \text{ m}) = 4.35207 \text{ N}$

Kinetic energy, $K = \frac{1}{2}mv^2$
 Elastic potential energy, $U = \frac{1}{2}kx^2$
 Total mechanical energy,
 $E = K + U = \frac{1}{2}kA^2$

Objects over a Pulley

$m_1 = 5 \text{ kg}$
 $m_2 = 3 \text{ kg}$
 $h_1 = 4 \text{ m}$

Animation speed
 Animate << Stop >> Reset
 $t = 1.66 \text{ s}$

Speed, $v_1 = 4.067 \text{ m/s}$
 Speed, $v_2 = 4.067 \text{ m/s}$ Max speed, $v_{\text{max}} = 4.4272 \text{ m/s}$

☒ Show analysis before m_1 hits the ground

$K_1 = 41.351 \text{ J}$ $U_{1i} = 196.00 \text{ J}$ $U_{2i} = 30.595 \text{ J}$
 $K_2 = 24.811 \text{ J}$ $U_2 = 99.243 \text{ J}$
 $\Delta K = 66.162 \text{ J}$ $\Delta U = -66.162 \text{ J}$

☐ After m_1 hits the ground and before m_2 comes to rest

Kinetic energy, $K = \frac{1}{2}mv^2$
 Gravitational potential energy, $U = mgy$

Impulse, Collision Time, and Impulsive Force

$m = 0.65 \text{ kg}$ $u = 8.2 \text{ m/s}$ Contact time, $\Delta t = 0.22 \text{ s}$

Animate Stop Reset Animation speed

$t = 0.932 \text{ s}$

Impulse, $I = \Delta p = 10.66 \text{ Ns}$
 Maximum impulsive force, $F_{max} = 87.083 \text{ N}$

☒ Show average impulsive force
 Average impulsive force, $F_{av} = 48.454 \text{ N}$
 Contact time, $\Delta t = 0.22 \text{ s}$

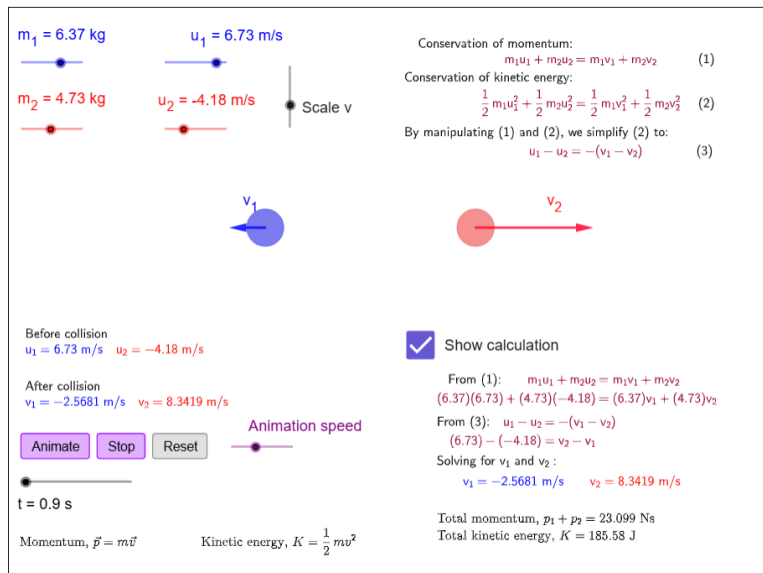
Impulsive force, $F \text{ (N)}$

Impulsive force, $F = 74.3 \text{ N}$

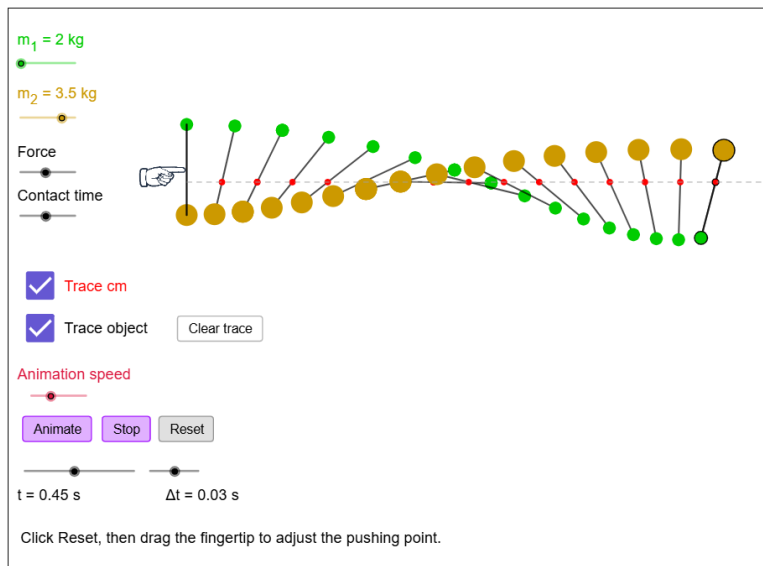
Time, $t \text{ (s)}$

To observe the impulsive force more closely during the collision, click Stop before the collision, reduce the animation speed, then click Animate again.

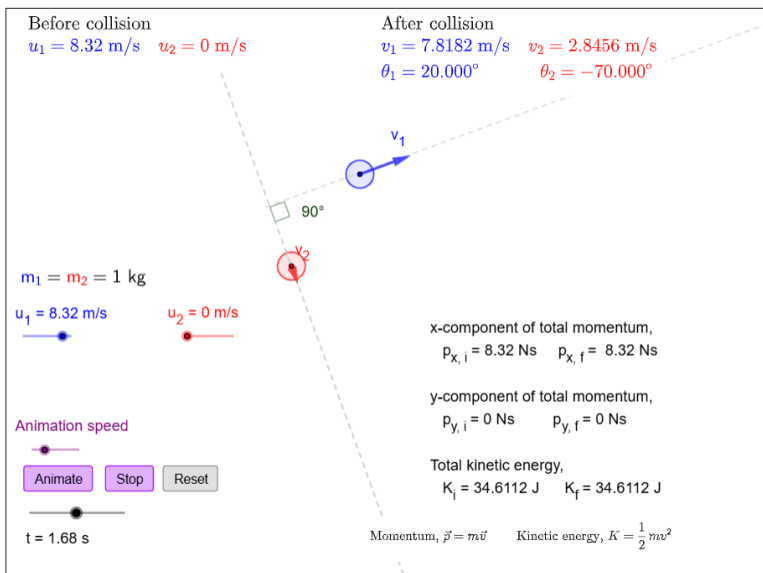
Conservation of Momentum and Kinetic Energy in One-Dimensional Elastic Collisions



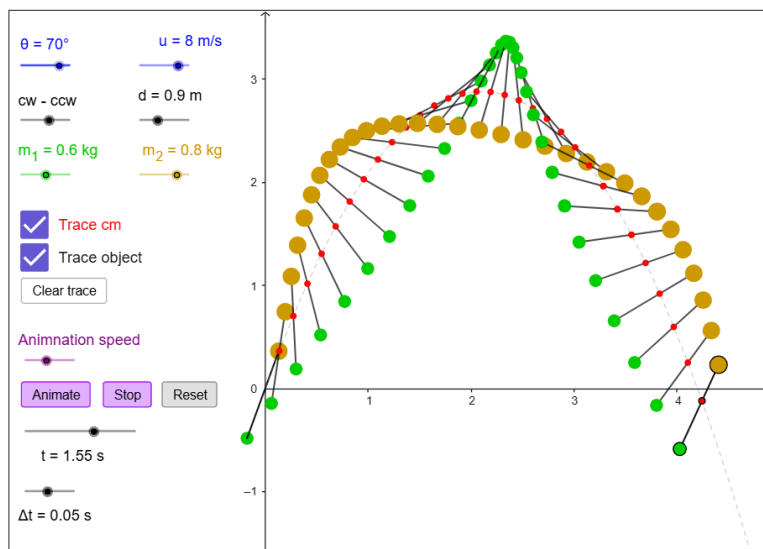
Translational vs. Rotational Motion of an Extended Object



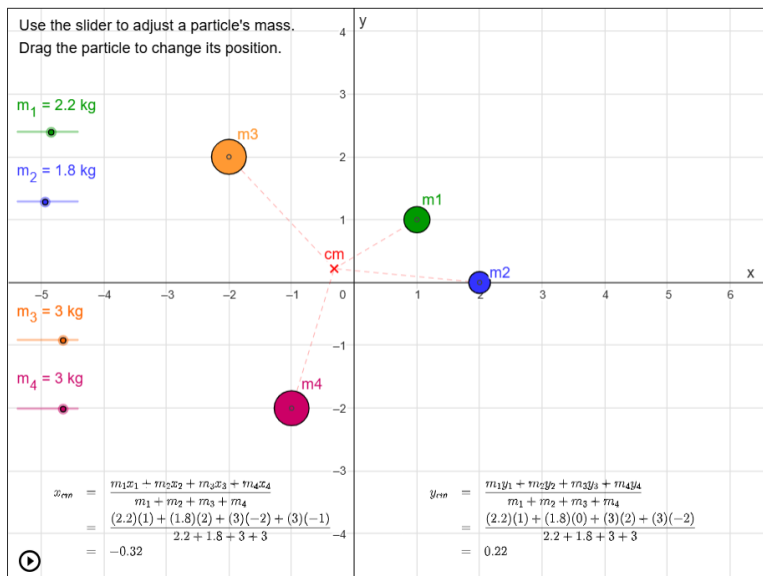
Conservation of Momentum and Kinetic Energy in Two-Dimensional Elastic Collisions



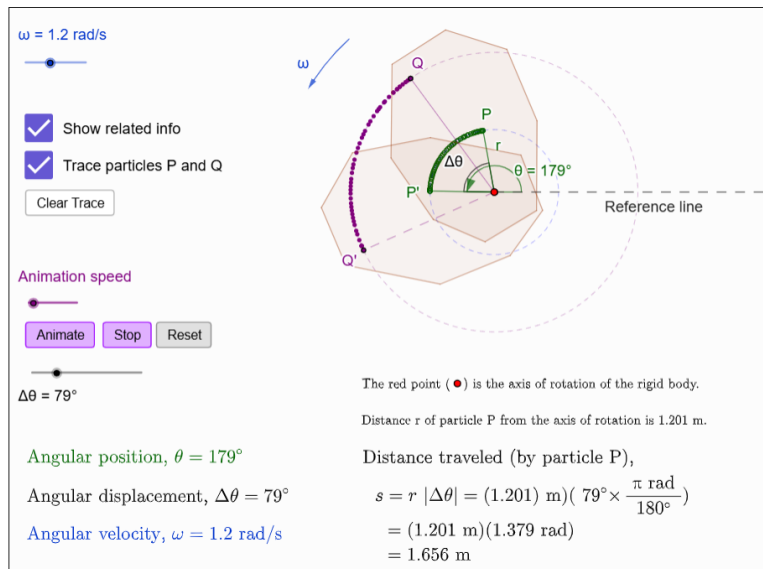
Projectile Motion of an Extended Object



Calculating the Center of Mass for a System of Particles



Rotational Kinematics of a Rigid Body



Rotational Kinematics under Constant Angular Acceleration

$\theta_0 = 60^\circ$ $60^\circ = 60^\circ \times \frac{\pi \text{ rad}}{180^\circ} = 1.0472 \text{ rad}$

 $\omega_0 = -3.3 \text{ rad/s}$

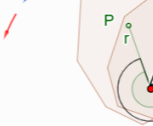
 $\alpha = 0.8 \text{ rad/s}^2$

☐ Comes to stop
☒ Show related info ☐ Trace P

Animation speed



 $t = 5.98 \text{ s}$



Angular displacement, $\Delta\theta = -5.4298 \text{ rad}$
 Distance traveled (by particle P),
 $s = r \Sigma|\Delta\theta| = (1.774)(6.806 + |-5.43| + |-6.806|)$
 $= (1.774)(8.183) = 14.519 \text{ m}$

$|\Delta\theta| = 6.806 \text{ rad at } t = 4.125 \text{ s when the rotating body stops momentarily.}$

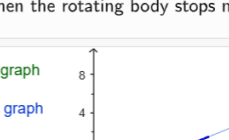
Angular position, $\theta = -4.3826 \text{ rad}$

Angular velocity, $\omega = 1.484 \text{ rad/s}$

Angular acceleration, $\alpha = 0.8 \text{ rad/s}^2$

$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$
 $\omega = \omega_0 + \alpha t$

☒ θ -t graph
☒ ω -t graph
☒ α -t graph



Rotational Inertia of Particles about an Axis

The screenshot shows a 3D coordinate system with x, y, and z axes. A triangular mass distribution is shown in the xz-plane. The vertices of the triangle are at (0,0,0), (4,0,0), and (0,3,0). The mass is distributed such that the center of mass is at (0,0,0). The axes are labeled with values 2 and 4. The triangle is shaded in light blue. The axes are labeled x, y, and z. The origin is labeled 0. The axes are labeled with values 2 and 4. The triangle is shaded in light blue. The axes are labeled x, y, and z. The origin is labeled 0. The axes are labeled with values 2 and 4.

Rotates about the x axis ☒

Rotates about the y axis ☐

Rotates about the z axis ☐

Animation speed

Animate Stop Reset

56°

$m_1 = 1 \text{ kg}$

$m_2 = 1 \text{ kg}$

$m_3 = 3 \text{ kg}$

Rotational inertia of a system of particles about an axis, $I = \sum m_i r_i^2$

$I_x = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 = (1)(0)^2 + (1)(0)^2 + (3)(4)^2 = 48 \text{ kg}\cdot\text{m}^2$

$I_y = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 = (1)(0)^2 + (1)(3)^2 + (3)(3)^2 = 36 \text{ kg}\cdot\text{m}^2$

$I_z = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 = (1)(0)^2 + (1)(3)^2 + (3)(5)^2 = 84 \text{ kg}\cdot\text{m}^2$

Rolling Motion Without Slipping

Radius, $r = 2 \text{ m}$

Ground

$x_{CM} = 7.7144 \text{ m}$

Animation speed

☒ Show the trace of the center of mass

$221^\circ = 221^\circ \times \frac{\pi \text{ rad}}{180^\circ} = 3.8572 \text{ rad}$

$\Delta\theta = 221^\circ$

Horizontal displacement of center of mass, $x_{CM} = 7.7144 \text{ m}$

Arc length, $s = r \Delta\theta = (2 \text{ m})(221^\circ \times \frac{\pi \text{ rad}}{180^\circ})$

$= 7.7144 \text{ m}$

Newton's Second Law in Rotational and Linear Motion

$R = 0.35 \text{ m}$
 $M = 4.8 \text{ kg}$
 $m = 3 \text{ kg}$
 $h = 3 \text{ m}$
 $t = 0.54 \text{ s}$

Animation speed
 [Animate] [Stop] [Reset]

$d = 0.7938 \text{ m}$

Rotational inertia,
 $I = \frac{1}{2} M R^2 = 0.294 \text{ kg}\cdot\text{m}^2$

☒ Show forces
 $mg = 29.400 \text{ N}$
 $T = 13.067 \text{ N}$

☒ Show angular quantities
 $\alpha = 15.556 \text{ rad/s}^2$
 $\omega = 8.4000 \text{ rad/s}$

☒ Show linear quantities
 $a = 5.4444 \text{ m/s}^2$
 $v = 2.9400 \text{ m/s}$
 $d = 0.79380 \text{ m}$

$\tau_{net} = I\alpha$ $F_{net} = ma$ $a = R\alpha$
 $\omega = \omega_o + \alpha t$ $v = u + at$ $v = R\omega$ $d = ut + \frac{1}{2} at^2$

Linear and Angular Quantities of a Rotating Rigid Body

$r = 13.2 \text{ m}$
 $\omega_o = 0.5 \text{ rad/s}$
 $\alpha = -0.07 \text{ rad/s}^2$
 Animation speed
 t = 2.58 s
 Radius, $r = 13.2 \text{ m}$
 Angular velocity, $\omega = 0.31940 \text{ rad/s}$
 Angular acceleration, $\alpha = -0.07 \text{ rad/s}^2$
 Linear velocity, $v = r\omega = 4.2161 \text{ m/s}$
 Centripetal acceleration, $a_c = r\omega^2 = 1.3466 \text{ m/s}^2$
 Tangential acceleration, $a_t = r\alpha = -0.92400 \text{ m/s}^2$
 Total acceleration, $a = \sqrt{a_c^2 + a_t^2} = 1.6331 \text{ m/s}^2$
 Angle ϕ between $\vec{a}$ and $\vec{a}_c = 34.457^\circ$
 $\omega = \omega_o + \alpha t$ $v = u + a_t t$

Bridge with Hinge and Cable Support

The diagram shows a horizontal beam of length 5 m attached to a wall by a hinge. A cable is attached to the wall at a height of 12 m and to the beam at a distance of 5 m from the hinge. The cable makes an angle of 20° with the horizontal. A mass m is hanging from the beam at a distance of 1 m from the right end. The weight of the beam is Mg and the weight of the mass is mg . The reaction force at the hinge is R . The angle between the beam and the horizontal is 51.0832°.

Parameters and values:

- L (m): 8
- M (kg): 2000
- m (kg): 1000
- $\alpha = -20^\circ$

Controls:

- Show force: ☒
- Scale force:

Calculated values:

- $T = 35507 \text{ N}$
- $R_x = 11511 \text{ N}$
- $R_y = -4189.5 \text{ N}$

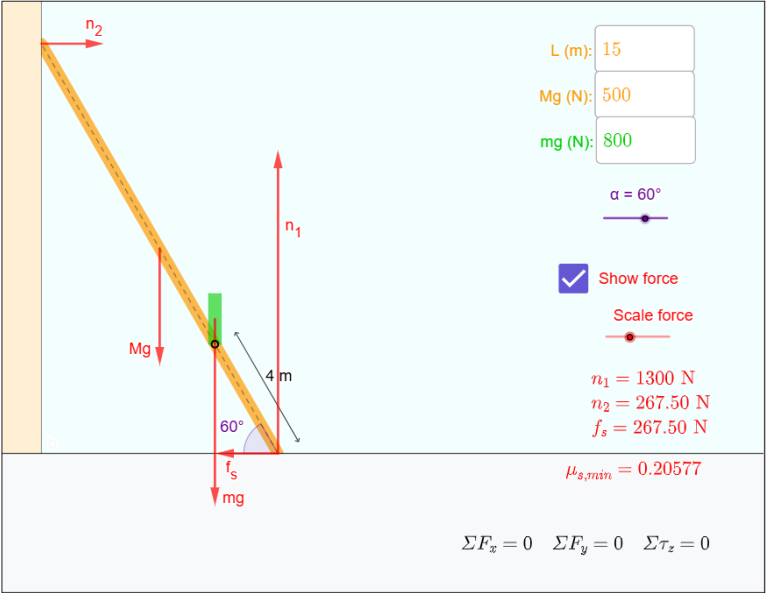
Instructions:

Drag the tiny hollow circles to change: the position of the object m and the connecting points of the cable to the bridge and the wall.

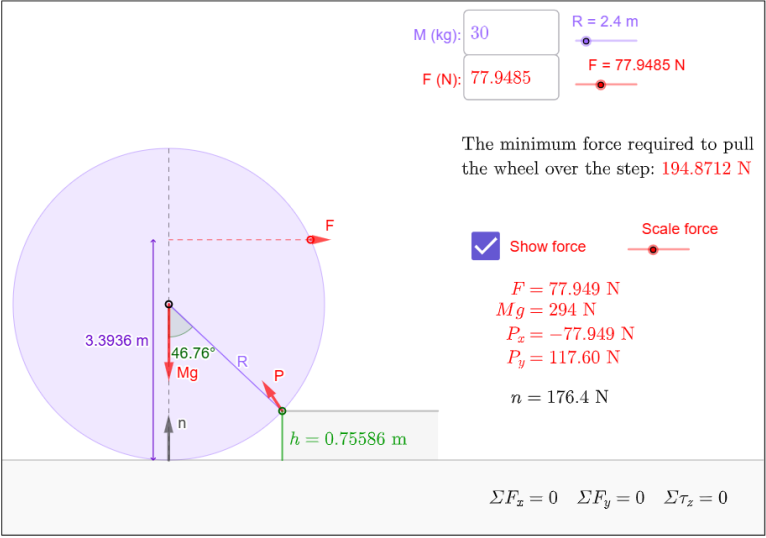
Equations:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma \tau_z = 0$$

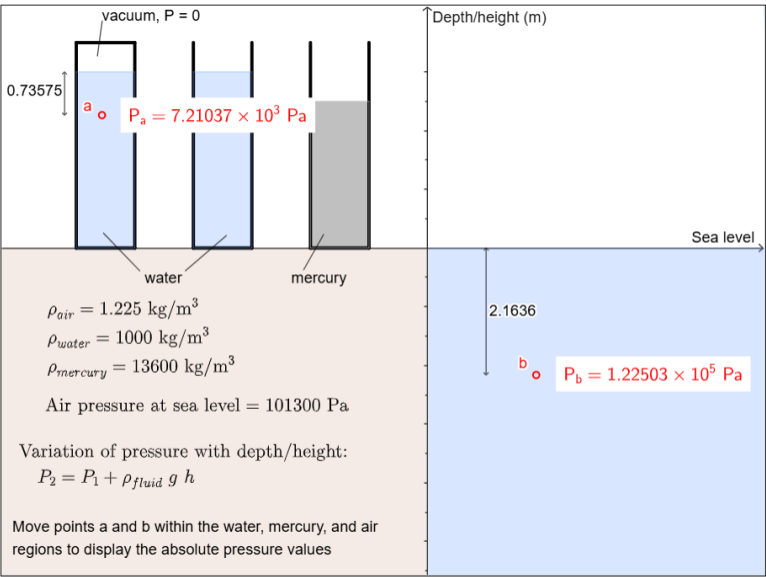
Ladder Leaning Against a Frictionless Wall



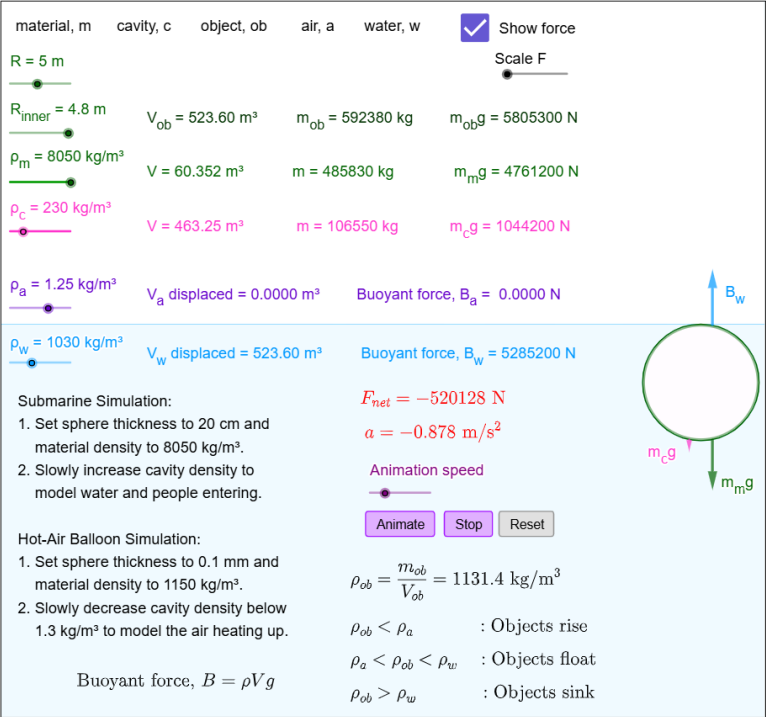
Minimum Force Required to Pull a Wheel Over a Step



Variation of Pressure with Depth and Height



Buoyant Force – Why Objects Sink, Float, or Rise



Buoyancy and Force Equilibrium

